MIL-T-22361(Wep)
8 March 1960
SUPERSEDING
NAVORD OSTD 53
30 June 1944

MILITARY SPECIFICATION

THREAD COMPOUND; ANTISEIZE, ZINC DUST-PETROLATUM

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers one grade of zinc dust-petrolatum antiseize thread compound.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

TT-P-460 - Pigment, Zinc-Dust (Metallic-Zinc-Powder), Dry

VV-P-236 - Petrolatum, Technical

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8030

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3. REQUIREMENTS

3.1 Materials. - The compound shall be a smooth, homogeneous mixture of zinc dust and petrolatum.

3.2 Composition. - The composition shall be as specified in Table I.

TABLE I

Composition

Ingredients	Percent by weight	
	(min)	(max)
Petrolatum	57	59
Zinc Dust	41	43

3.3 Properties of the compound. -

3.3.1 Penetration. - The normal worked penetration shall not be less than 160 nor more than 240.

3.3.2 Stability. - The compound shall show no separation when a 50 gram sample is centrifuged at room temperature for 1/2 hour at a relative centrifugal force of 500 ± 25.

3.4 Properties of the petrolatum ingredient. -

3.4.1 The petrolatum shall conform to the requirements of Specification VV-P-236.

3.5 Properties of the zinc dust ingredient. -

3.5.1 The zinc dust shall conform to the requirements of Specification TT-P-460, Type I.

3.6 Workmanship. - The compound shall be a smooth, homogeneous product free from abrasives, filler and impurities. There shall be no separation of the mixture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the

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Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 General. - Inspection requirements outlined in Method 9601 of Federal Test Method Standard No. 791 shall be incumbent on all suppliers.

4.1.2 Report of analysis. - The contractor shall furnish to the inspector reportecertified by affidavit, showing the composition of each batch in the shipment.

4.2 Acceptance tests. - The acceptance tests shall consist of the following tests as described in 4.3, 4.4, and 4.5.

4.2.1 Acceptance test samples. - The inspector shall take at random for acceptance tests at least five 1/2 pound tubes or three one pound containers from each lot under inspection. For users who obtain the compound in large containers, three one pound samples shall be taken from separate containers for acceptance tests; however, care should be exercised that the receptacles for the acceptance samples are clean and dry. A lot shall consist of 500 pounds or less of compound, manufactured at one time from one batch of compound, submitted for inspection at the same time and place. If more than one batch of compound is included in the lot, a random sample shall be taken from each batch. A batch is defined as the end product of all raw material blended in a single operation.

4.2.2 Sampling for inspection of filled containers. - A random sample of filled containers shall be selected from each lot offered for Government inspection in accordance with Standard MIL-STD-105 at Inspection Level I, and Acceptable Quality Level (AQL) of 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements established by MIL-STD-290.

4.3 Examination of product. - The sample of the compound shall be inspected visually and a spatula-full taken up shall be worked with the spatula on a glass surface. After working, the compound shall be spread with a straight edge and observed for homogeneity as indicated by a smooth surface, and also for the absence of abrasives and impurities.

4.4 Penetration. - The normal worked penetration of the compound shall be determined in accordance with Federal Test Method Standard No. 791, Method No. 311.5.

4.5 Stability. - The stability of the compound shall be determined by placing 50 grams of the compound in each of two 50-ml. centrifuge tubes. The centrifuge and the method of calculating rpm are described in Federal Test Method Standard No. 791, Method No. 3003.5.

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4.6 Rejection and retest. - Failure of any compound sample to conform to the requirements of this specification shall be cause for rejection of the lot represented. Rejected compound shall not be resubmitted for acceptance without full particulars being furnished to the procuring activity concerning previous rejection and measures taken to overcome the defects.

4.7 Packaging. - The inspector shall ascertain that packaging of the compound conforms to the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing. - The packaging and packing of the compound shall be Levels A, B, or C as specified in the contract or order (see 6.4) in accordance with MIL-STD-290. The quantity of compound in pounds and type and capacity of the container shall be as specified in the contract or order (see 6.4). The container or any component thereof (closure, lining, etc.) shall not interact or alter in any way so as to affect adversely the purity or quality of the container contents. All containers shall be new and free from contaminations.

5.3 Marking. - Marking shall be in accordance with MIL-STD-290.

6. NOTES

6.1 Intended use. - The intended use of this compound is to prevent seizing during assembly or disassembly of threaded or unthreaded components fabricated from aluminum or its alloys, engaged with components fabricated from similar or dissimilar metals. It is also intended to provide corrosion protection to the metal surfaces.

6.2 Care shall be taken against the application of too heavy a coating to the components. A thin coating is all that is required to prevent seizing. In the case of blind holes, the application of an excessive amount of the compound may prevent proper seating of the components. Under low temperature conditions this compound hardens and is difficult to apply. It should be kept at room temperature for twenty-four to forty-eight hours prior to use.

6.3 Due to the high evaporation rate of the petrolatum constituent, this compound is not suitable for use on the threaded or unthreaded components of such equipment as optical instruments where the vapors may adversely effect associated components e.g., lenses, prisms and other optical elements.

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6.4 Ordering data. - Requisitions, contracts, or orders should specify the following:

(a) Title, number, and date of this specification
(b) Quantity desired
(c) Applicable levels of packaging and packing
(d) Type and capacity of containers required

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any rights or person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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